

Cryptography and Network Security

Factorization Methods

Today's Lecture : Factorization Methods

- Fermat's Factorization method
- Pollard P-1 method
 - This method discovers a prime factor p of an integer n whenever $p-1$ has only small prime factors.
 - It was derived using Fermat's theorem.
 - We are not going to check the proof.

Fermat's Factorization method

- **Prime factorization is known as a way to crack the RSA cryptosystem code.**
- Currently, most of the best modern factoring algorithms are based on the idea behind Fermat's method of factorization.
- Fermat's factorization method uses that fact that **any number can be expressed as the difference between two squares.**
- Fermat's Factorization is an integer factorization technique **used to express an odd composite number as a difference of two squares.**

Fermat's Factorization method

- Basic Idea:
- Any odd integer N can be written as:
- $N = a \cdot b$ (This works well when a and b are close)

- **Formula :**

$$N = a^2 - b^2 = (a - b)(a + b)$$

$$b^2 = a^2 - N$$

Fermat's Factorization method

- **Algorithm Steps:**

1. Choose $a = \lceil \sqrt{N} \rceil$ where $\lceil \cdot \rceil$ is the ceiling function (round up)
2. Compute $b^2 = a^2 - N$
3. If b^2 is a perfect square, stop.
4. Otherwise, increment a and repeat.
5. Factors are $(a - b)$ and $(a + b)$

Fermat's Factorization method

Example: Factorize 5959

Step 1: Choose $a = \lceil \sqrt{N} \rceil$

$$\sqrt{5959} \approx 77.2, \text{ so } a = 78$$

Step 2: Compute $b^2 = a^2 - N$

$$\text{for } a = 78 ; a^2 = 78^2 = 6084$$

$$b^2 = 6084 - 5959 = 125 \text{ (not a perfect square)}$$

Step 3: increment a

$$a = 79 ; a^2 = 79^2 = 6241$$

$$b^2 = 6241 - 5959 = 282 \text{ (not a perfect square)}$$

$$a = 80 ; a^2 = 80^2 = 6400$$

$$b^2 = 6400 - 5959 = 441 \text{ (perfect square)}$$

$$b = 21$$

Step 4: compute factors

$$\text{Now } b = 21, a = 80$$

$$N = (a - b)(a + b)$$

$$5959 = (80 - 21)(80 + 21)$$

$$= 59 \times 101$$

Practice Exercises

Example1: Factorize 8051

Example2: Factorize 3233

Example3: Factorize 119143

Fermat's Factorization method

- The security of RSA algorithm depends upon the integer N , which is the multiple of two large prime numbers p and q .
- **Fermat's factorization method factors N into p and q very efficiently if p and q share half of their leading bits. In other words, if the gap between p and q is below the square root of p .**

Some terminology

- **What is B-smooth number?**

- Imagine you're breaking a number down into its smallest building blocks these are **prime numbers** (like 2, 3, 5, 7, 11, ...).
- **Smooth** means "**no big prime chunks**" in that breakdown.
- **B-smooth** means: when you break the number into primes, **all those primes are $\leq B$** .

Example :

We take **1,620** and factor it:

$$1620 = 2 \times 810$$

$$810 = 2 \times 405$$

$$405 = 3 \times 135$$

$$135 = 3 \times 45$$

$$45 = 3 \times 15$$

$$15 = 3 \times 5$$

Putting it together:

$$1620 = 2 \times 2 \times 3 \times 3 \times 3 \times 3 \times 5$$

Check if it's B-smooth for $B = 5$

We look at all primes in the factorization:

{2,3,5}

Are any of these primes **greater than 5**?

2 > 5? No

3 > 5? No

5 > 5? No (5 is equal to 5, and that's allowed; " $\leq B$ " means less than or equal to B.)

So all primes are ≤ 5 .

Thus **1620 is 5-smooth**.

What about $B = 3$?

Then we check: prime factors are

{2, 3, 5}.

5 is greater than 3?

Yes ($5 > 3$). **So 1620 is NOT 3-smooth** because it has a prime factor 5 which is bigger than 3.

Some terminology

- **General rule for B-smooth numbers**

- If $B=5$, then the prime factors allowed are only **2, 3, 5**.
- So any 5-smooth number will look like:

$$2^a \times 3^b \times 5^c$$

- where a, b, c are whole numbers ≥ 0 .

- **Example:**

- $45 = 3^2 \times 5^1 \rightarrow$ 5-smooth ✓
- $50 = 2^1 \times 5^2 \rightarrow$ 5-smooth ✓
- $14 = 2 \times 7 \rightarrow$ NOT 5-smooth (has prime $7 > 5$)

Some terminology

- **What is B-Power smooth number?**

- **B-smooth** means all prime factors are $\leq B$.
- **B-power smooth** is stricter: Every prime *power* in the number's factorization must be $\leq B$.

Example:

$$720 = (2^4 \times 3^2 \times 5^1)$$

Prime powers: $2^4 = 16$, $3^2 = 9$, $5^1 = 5$

Is 720 5-smooth?:

Yes, primes {2,3,5} all ≤ 5 .

Is 720 5-power smooth?

No : because $16 > 5$ and $9 > 5$.
We need all prime powers ≤ 5 for that.

- For 720, largest prime power is $2^4 = 16$.
- So 720 is 16-power smooth (because $16, 9, 5 \leq 16$).
- It's also 17-power smooth, 18-power smooth, etc.

Calculate Following?

- Check if following numbers are 7 smooth or not?
 - 252
 - 99
- Whether above numbers are 7 – powersmooth or not?

Pollard's $p - 1$ factorization algorithm

1. Given a number n . Initialize $a = 2$ and a number $B = 2$
2. Loop Until factor found
 1. Compute $a = (a^{B!}) \bmod n$
 2. Compute $d = \text{GCD}(a - 1, n)$
 3. if $1 < d < n$ then return d , $\rightarrow$ **Stop**
 4. If not, increase B by 1 and repeat ($B = B+1$)
3. Other factor, $d' = n/d$
4. If d' is not prime , then $n = d'$ and goto 1 else d and d' are two prime factors.

Example:

Let $n = 10001$. Let's start with $a = 2$. Then clearly $\gcd(2, 10001) = 1$, so we proceed into the loop. We first compute $a^{2!} = 2^2 = 4$. Then

$$\gcd(a^{2!} - 1, n) = \gcd(3, 10001) = 1$$

so we continue. Now $a^{3!} = (a^{2!})^3 = 4^3 = 64$, and

$$\gcd(a^{3!} - 1, n) = \gcd(63, 10001) = 1$$

so we continue. Next $a^{4!} = (a^{3!})^4 = 64^4 \equiv 5539 \pmod{10001}$, and

$$\gcd(a^{4!} - 1, n) = \gcd(5538, 10001) = 1$$

so we continue. Next $a^{5!} = (a^{4!})^5 = 5539^5 \equiv 7746 \pmod{10001}$, and

$$\gcd(a^{5!} - 1, n) = \gcd(7745, 10001) = 1$$

so we continue. Next we have $a^{6!} = (a^{5!})^6 = 7746^6 \equiv 1169 \pmod{10001}$, and

$$\gcd(a^{6!} - 1, n) = \gcd(1168, 10001) = 73.$$

We've run into a gcd that is bigger than 1 and not equal to $n = 10001$, so jackpot! 73 must be a prime factor of n . Then we can compute quickly that $10001/73 = 137$, so

$$10001 = 73 \cdot 137$$

Practice Exercise:

1. Let $n=323$. Lets Start with $a = 2$
2. Let $n=8051$. Lets Start with $a = 3$

Example

Step 1: Initialization

- $n = 1403$
- $a = 2$ (starting base)
- $j = 2$ (iteration counter, equivalent to i in algorithm)

Step 2: Iteration 1 ($j = 2$)

Compute:

$$a = a^j \mod n = 2^2 \mod 1403 = 4$$

Check GCD:

$$d = \gcd(a - 1, n) = \gcd(4 - 1, 1403) = \gcd(3, 1403) = 1$$

Since $d = 1$, no factor found.

Increment j :

$$j = 2 + 1 = 3$$

Step 3: Iteration 2 ($j = 3$)

Compute:

$$a = a^j \mod n = 4^3 \mod 1403$$

$$4^3 = 64$$

($64 < 1403$, so $64 \mod 1403 = 64$)

Check GCD:

$$d = \gcd(64 - 1, 1403) = \gcd(63, 1403) = 1$$

No factor.

Increment j :

$$j = 3 + 1 = 4$$

Step 4: Iteration 3 ($j = 4$)

Compute:

$$a = a^j \mod n = 64^4 \mod 1403$$

Step-by-step modulo:

1. $64^2 = 4096$

$$4096 \div 1403 : \text{quotient } 2, \text{ remainder } 4096 - 2 \times 1403 = 4096 - 2806 = 1290$$

$$\text{So } 64^2 \equiv 1290 \pmod{1403}$$

2. Square again: $1290^2 = 1,664,100$

Divide by 1403:

$$1403 \times 1186 = 1403 \times (1000 + 186)$$

$$= 1,403,000 + (1403 \times 186)$$

$$1403 \times 186 = 1403 \times 200 - 1403 \times 14 = 280,600 - 19,642 = 260,958$$

$$\text{Sum: } 1,403,000 + 260,958 = 1,663,958$$

$$\text{Remainder: } 1,664,100 - 1,663,958 = 142$$

$$\text{So } 64^4 \equiv 142 \pmod{1403}$$

Check GCD:

$$d = \gcd(142 - 1, 1403) = \gcd(141, 1403) = 1$$

No factor.

Increment j :

$$j = 4 + 1 = 5$$

Step 5: Iteration 4 ($j = 5$)

Compute:

$$a = a^j \mod n = 142^5 \mod 1403$$

Step-by-step:

1. $142^2 = 20,164$

$$1403 \times 14 = 19,642$$

$$\text{Remainder: } 20,164 - 19,642 = 522$$

$$\text{So } 142^2 \equiv 522$$

2. $142^4 = (142^2)^2 \equiv 522^2$

$$522^2 = 272,484$$

$$1403 \times 194 = 1403 \times 200 - 1403 \times 6 = 280,600 - 8,418 = 272,182$$

$$\text{Remainder: } 272,484 - 272,182 = 302$$

$$\text{So } 142^4 \equiv 302$$

3. $142^5 = 142^4 \times 142 \equiv 302 \times 142$

$$302 \times 142 = 302 \times 100 + 302 \times 42 = 30,200 + 12,684 = 42,884$$

$$1403 \times 30 = 42,090$$

$$\text{Remainder: } 42,884 - 42,090 = 794$$

$$\text{So } a = 794$$

Check GCD:

$$d = \gcd(794 - 1, 1403) = \gcd(793, 1403)$$

Use Euclidean algorithm:

- $1403 - 793 = 610$
- $793 - 610 = 183$
- $610 - 3 \times 183 = 610 - 549 = 61$
- $183 - 3 \times 61 = 0$

So $\gcd = 61$.

Step 6: Factor found

Since $1 < 61 < 1403$, we have a nontrivial factor:

$$p = 61$$

Other factor:

$$q = 1403/61 = 23$$

Step 7: Verification

$$61 \times 23 = 1403 \quad \checkmark$$